

- 1 The photograph shows a germinating pea seed.



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Magnification $\times 2.0$

The embryo in a seed will start to grow when temperature and moisture conditions are favourable.

The seed contains a food supply which is broken down by enzymes. This provides substrates for respiration.

(a) Describe an experiment to investigate the effect of temperature on the rate of respiration of germinating pea seeds.

(5)

[illegible]

(b) State two variables, other than the independent variable, that could affect this experiment.

(2)

(c) Name one of the variables you have identified in (b).

Variable

(i) State how this variable could be controlled.

(1)

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(ii) Suggest what effect not controlling this variable could have on the results.

(1)

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(d) Explain how a lack of oxygen could affect the growth of seedlings after germination.

(3)

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(Total for Question 1 = 12 marks)